Personal Cybersecurity Plan

Case Study: Home Network Security Assessment

1. Devices, Services, Accounts and Their Security

The following devices and services represent a typical home user environment subject to assessment:

Device / Service	Current Security Status
Wi-Fi Router	Default password unchanged; no guest network; password freely shared
Mobile Phone	4-digit screen lock; SIM PIN enabled; manual updates; many unused apps
Laptop	Outdated antivirus; automatic updates enabled; device password set; no guest account
Networked Devices (IoT)	Dishwasher, printer, and set-top box connected to the main Wi-Fi network; no dedicated IoT network segment; firmware updates performed manually or not at all
Social Media Accounts	2FA enabled via authenticator app on all accounts
E-commerce Account	2FA enabled via authenticator app
Email Accounts	2FA enabled on all accounts
Passwords	All unique; password manager in use
Child Devices	Parental controls active
Saved Payment Info	Payment card stored on one shopping platform only

2. Potential Threats

Device / Service	Potential Threat
Wi-Fi Router	Unauthorised access to the network; data interception
Laptop	Malware infection; data theft via outdated antivirus
Mobile Phone	Vulnerabilities introduced by unused or outdated apps
Social Media / Email	Phishing attacks; account compromise
Password Manager	If the primary account is compromised, all stored passwords are exposed
Child Devices	Exposure to inappropriate content or contact with unknown individuals
Saved Payment Info	Financial fraud if the associated account is hacked
Networked Devices	Unpatched firmware creates exploitable vulnerabilities;

(IoT)	compromised IoT device can serve as a pivot point to attack other devices on the same network
Shared Device Use	Unauthorised access to sensitive data by third parties

3. Threat Mitigation Plan

- Create a dedicated IoT network segment (separate SSID/VLAN) for smart appliances such as the dishwasher, printer, and set-top box — isolating them from devices that handle sensitive data
- Enable automatic firmware updates on all IoT devices where supported; for devices without auto-update (e.g., dishwasher), set a monthly reminder to check the manufacturer's website for patches
- Change the Wi-Fi router default password and configure a separate guest network
- Remove unused mobile applications to reduce the attack surface
- Extend the mobile screen lock from 4 to 6+ digits or enable biometric authentication
- Update antivirus software on the laptop and ensure real-time protection is active
- Enable a guest account on the laptop for use by third parties
- Avoid reusing usernames across services; diversify credentials
- Periodically rotate passwords, even when using an auto-generating password manager
- Enable WPA2/WPA3 encryption on the Wi-Fi router
- Review browser privacy settings and consider using privacy-focused extensions
- Avoid storing payment information on shopping platforms; use secure, tokenised payment options

4. Additional Security Recommendations

- Enable account security alerts for suspicious sign-in activity
- Consider an alternative, locally-stored password manager (e.g., Bitwarden)
- Subscribe to cybersecurity awareness resources and scam alert services
- Review device and account security settings on a monthly basis

Conclusion

This cybersecurity plan identifies and addresses realistic personal digital threats across a typical home user environment. It demonstrates awareness of device and account vulnerabilities and provides clear, actionable mitigation steps. The plan reflects core principles of risk assessment and personal security planning as applied in professional cybersecurity practice.